HAMLET: Adaptive Lesion-Graph Reasoning with Evidential Prediction for Multimodal Low-Dose CT Lung Cancer Assessment

Arpita Tiwari

Department of Electronics and Communication Engineering,

Indian Institute of Information Technology Nagpur, Nagpur 441108, India

dtea24ece004@iiitn.ac.in

Abstract

A screening chest CT is not one finding but a scene: several nodules of different size and density, old scarring, and a participant whose age and smoking history carry risk that no texture descriptor recovers. Dense whole-volume encoders flatten that scene; lesion graphs preserve it but reach it through a hard detector threshold and a neighbourhood rule fixed before training. HAMLET removes both constraints. Candidate lesions come from a query-based multi-resolution token bank and stay differentiable with respect to the examination-level objective; a relaxed binary structure learner, not spatial distance, decides which lesions communicate; clinical covariates enter as a node in the same graph; and a Dirichlet head returns a risk estimate with a vacuity score while a halting gate sets reasoning depth per examination. On a held-out NLST cohort of 2,068 examinations, HAMLET reaches AUROC 0.935 and ten-bin ECE 0.039, against 0.923/0.049 for a static lesion graph with the same evidential head and 0.914/0.083 for a dense multimodal encoder. Component replacement shows the mechanisms doing different work: representation and connectivity carry discrimination, the evidential objective carries calibration, and adaptive depth halves the reasoning rounds at unchanged accuracy.

Keywords: low-dose CT; lung cancer screening; lesion graph; graph structure learning; multimodal fusion; evidential deep learning; mixture of experts; adaptive computation; calibration.

1 Introduction

The National Lung Screening Trial (NLST) showed that LDCT screening lowers lung-cancer mortality in a high-risk population [1], and screening programmes built on that evidence now generate imaging volumes that outpace the radiology workforce available to read them. The modelling problem, however, is not simply one of throughput. A screening examination is a heterogeneous scene. It may contain a single solid nodule, several synchronous findings of different attenuation, post-inflammatory scarring, or diffuse parenchymal change; and the participant attached to that scene carries an age, a smoking history and a set of comorbidities that carry risk information no texture descriptor recovers. A useful system has to aggregate across that scene, use the non-imaging information, and then say something honest about its own confidence.

Three design decisions govern how well a system does this, and existing work tends to fix each one early. The first is how candidate findings are represented. Whole-volume encoders such as M3FM [4] process the scan uniformly and fuse clinical variables afterwards; the approach transfers well across tasks but never materialises a set of objects whose relationships can be examined. Candidate-based pipelines materialise that set, yet the candidates usually arrive from a separately trained detector through a thresholding step, so the examination-level loss cannot influence what was selected. The second decision is which candidates should influence one another. Lesion-graph methods typically define neighbourhoods by spatial proximity, using a k-nearest-neighbour or radius rule chosen before training, which ties relational reasoning to anatomical distance even when the informative relationship is morphological. The third is how confidence is produced. Softmax scores trained under cross-entropy are known to be poorly calibrated [18], and attaching an evidential head to an otherwise rigid backbone improves the reported calibration without making the representation itself adaptive.

HAMLET is an attempt to unfix all three within one architecture, and, more importantly for this manuscript, to test them separately. The framework combines a differentiable multi-resolution token bank, a learned sparse lesion graph, cross-scale message passing, a clinical covariate node inside that graph, token-level expert routing, a Dirichlet evidential head, and a halting gate that varies reasoning depth per examination. None of these primitives is new: feature pyramids [5], query-based set prediction [6], graph attention [7], learned discrete structure [8], relaxed binary sampling [9, 10], sparse expert layers [11], adaptive computation [13] and evidential classification [14] all have established antecedents, and we do not claim otherwise. The claim under test is narrower and, we would argue, more useful: that these mechanisms occupy distinct experimental roles, and that an evaluation can attribute discrimination, calibration and computational behaviour to different parts of the model rather than reporting one aggregate improvement.

Two internally trained comparators make that claim falsifiable. A dense multimodal baseline stands for whole-volume encoding followed by image–clinical fusion, and asks whether an explicit lesion set is worth constructing at all. A static graph with a Dirichlet head stands for candidate-based relational modelling with the predictive head already held constant, and asks the sharper question: once you have a relational model and an evidential objective, what remains attributable to adaptive candidate selection and learned connectivity? Without that second comparator, an improvement over a softmax-trained dense model would confound representation learning with a change of training objective, and the

reader would have no way to separate them. The reasoning throughout this paper follows the same chain for each mechanism — prior approach, its limitation, the baseline that embodies it, the HAMLET modification, its mathematical form, the experiment that isolates it, and what the observed change does and does not license.

2 Related work and comparative rationale

Rather than surveying the field chronologically, this section is organised by the methodological building blocks HAMLET modifies. For each block we state what prior work achieves, what it leaves unresolved for this task, and which internal experiment tests the corresponding change. Table 1 collects that mapping.

2.1 Screening endpoints and image-only prediction

Four tasks are routinely conflated in this literature, and separating them is a precondition for any honest comparison. A detector localises a nodule. A nodule classifier characterises an annotated candidate. An examination classifier aggregates evidence over a whole scan. A longitudinal risk model estimates the probability of a diagnosis within a stated future interval. These targets share image content but differ in their negatives, their denominators and their follow-up requirements.

Ardila et al. trained an end-to-end three-dimensional model using the current scan and, where available, a prior, reporting AUROC 0.944 on 6,716 NLST cases with an independent clinical reader comparison [2]. Sybil predicts future risk from a single LDCT without clinical inputs or radiologist marks, with one-year AUROC of 0.92 on NLST and 0.86 and 0.94 on two independent institutional cohorts [3]. Both establish that image-only prediction is strong on this data, and both are appropriate references for the imaging pathway. Neither is a leaderboard entry against which the number in this paper can be ranked, because the endpoint here is an examination-level cancer assessment whose horizon the present study does not define. That is a limitation of the present study, not a reason to ignore stronger work.

2.2 Candidate generation and detector warm-start

DeepSEED couples a three-dimensional encoder–decoder with region proposals and squeeze-and-excitation recalibration for pulmonary nodule detection [21], and the LUNA16 protocol provides the controlled setting in which candidate sensitivity is normally measured [22]. This lineage matters to HAMLET only at initialisation: the token decoder is warm-started on LUNA16 annotations before any NLST training. Recent benchmarking across DLCS, LUNA16, NLST-3D and LUNA25 makes the accompanying caution concrete, reporting nodule-classification AUROC between 0.71 and 0.90 for the same detection-informed model depending on which dataset it is evaluated on [24]. A single number from this family is therefore closer to a property of the evaluation corpus than of the method.

2.3 Multimodal fusion and what counts as a modality

M3FM organises multiscale CT together with 49 clinical data types in a question-answering framework spanning 17 screening tasks [4], and remains the closest published lineage to the present work. HAMLET takes a much smaller position: structured covariates are encoded into a node that participates in graph message passing, so clinical information conditions lesion representations before pooling rather than being combined with an image score afterwards. Gated multimodal units supply the fusion mechanism used by our dense comparator [27].

One clarification belongs here, because the point is easy to lose. Three CT resolutions are three views of one modality; a feature pyramid does not make a model multimodal. HAMLET is multimodal exactly when a clinical vector linked to the same participant and screening round enters the prediction path, which in this architecture happens at Eq. (5) and the node it creates. Section 3 therefore treats the clinical schema, the join and the temporal admissibility of each field as first-order scientific questions rather than reporting details, and Section 11 argues that an image-only versus clinical-only versus joint experiment is the single most valuable missing measurement in the study.

2.4 Lesion representation and learned graph structure

Feature pyramids give multi-scale representations [5], and DETR showed that learned queries with set-based matching can replace hand-designed assignment and non-maximum suppression [6]. HAMLET adapts both to a three-dimensional token bank that is supervised during warm-up and then optimised for examination-level prediction. The limitation of hard region-of-interest processing that matters here is the discrete selection boundary between localisation and classification, not a blanket claim that cropping is non-differentiable.

Graph attention learns how strongly neighbours influence a node [7]; learning *which* neighbours exist is a separate problem, addressed by Franceschi et al. through bilevel optimisation of discrete structure [8]. HAMLET's pairwise scorer with a relaxed Bernoulli sampler is a different parameterisation of that idea, and its advantage over a fixed neighbourhood is that it can connect morphologically informative candidates that are not spatially closest. Testing this requires holding candidate features and graph depth constant while swapping only the topology, which is what the fixed-kNN ablation does — with the edge-budget caveat discussed in Section 7.

2.5 Conditional computation and uncertainty

Sparse mixture-of-experts layers evaluate a subset of sub-networks per input [11, 12], and adaptive computation time varies the number of recurrent updates [13]. HAMLET applies the former at token level and the latter at examination

level. The two are frequently described together but do different work: routing selects a transformation, halting decides whether another round of reasoning happens at all. Neither measures uncertainty, and the halting gate takes the pooled representation as input, not an uncertainty scalar. We therefore call the mechanism representation-dependent adaptive halting throughout, and Eq. (17) is written so the reader can see why.

Evidential classification predicts Dirichlet parameters and derives a concentration-based uncertainty score from them [14]; trusted multi-view classification extends the idea by fusing evidence across views [15]. HAMLET fuses features before a single evidential output, which is a weaker but simpler design. A single deterministic evidence estimate is not a posterior over weights, and concentration does not by itself guarantee calibrated probabilities or reliable detection of distribution shift. This point is made repeatedly in the uncertainty-evaluation literature [16, 17]. A temperature-scaled softmax comparator [18] is therefore a necessary control, and it is listed in Table 1 and reported in Section 11 as the next experiment we intend to run.

3 Data, modalities and provenance

3.1 Primary cohort and the unit of analysis

NLST is the primary dataset for training and held-out assessment in both source versions, and it remains the declared source here. Precision about the unit of analysis matters. The trial randomised 26,722 participants to the LDCT arm [1], and NCI's imaging catalogue describes on the order of 75,000 screening examinations with multiple reconstructions per examination [33]. It is tempting to quote "$>$26,000 scans" for NLST, but that conflates the participant count with an image count; that entry is not retained.

We use a patient-level 70/10/20 split stratified by outcome and acquisition site, with a held-out set of 2,068 examinations containing 1,019 cancer-positive and 1,049 cancer-negative examinations. The implied prevalence is 49.27%, which is an order of magnitude above screening prevalence and identifies the test set as an outcome-enriched case–control sample rather than a screening-representative one. This is a legitimate design for model comparison and an inadequate one for calibration or workload claims, and the distinction is carried through every subsequent section. Per-partition counts, the number of unique test participants, and the sampling rule that produced the enrichment are reported in Supplementary Section S1; they cannot be reconstructed by multiplying examination totals by split fractions, and repeated screens or reconstructions can inflate positive records without adding independent positive participants.

The endpoint is described only as a confirmed cancer outcome. That is not sufficient to determine whether the target is cancer present at the index screen or cancer diagnosed within a defined interval afterwards. We therefore use the term *examination-level cancer assessment* and make no one-year risk claim. A reproducible label definition must state index date, diagnosis date, prediction horizon, minimum negative follow-up, censoring and the handling of repeated screens; and a cancer-negative examination must not be described as containing a histologically benign nodule, since the trial's negative definition is an outcome, not a pathology report.

3.2 Auxiliary image resources and the warm-up stage

LIDC-IDRI contains 1,018 thoracic CT studies with annotations from four readers [23]; LUNA16 is a curated subset of that resource with 888 scans and 1,186 reference nodules [22]. They are nested, so their sizes cannot be added, and their reference standard, radiologist suspicion, is not interchangeable with a confirmed NLST outcome.

We use 50 epochs of token-decoder warm-up on LUNA16 candidates. That stage is image-only supervised initialisation, as Fig. 1A makes explicit. CT volumes provide the input and candidate annotations the supervision; no clinical record is involved, and nothing but learned weights crosses into the NLST stage. Stating this plainly removes an ambiguity that could otherwise be read as a claim that pre-training was itself multimodal. No non-LUNA16 LIDC-IDRI scans or ratings enter training, so LIDC-IDRI functions here as the parent resource of LUNA16 rather than as a second pre-training corpus.

3.3 Clinical variables, linkage and temporal admissibility

The source manuscripts name age and smoking history, and one adds spirometry. The NLST participant dictionary does contain the corresponding fields, including smoking status at randomisation and pack-years [34]. Those are verified *candidate* source fields; they are not evidence that these exact columns entered the trained model, and spirometry is not retained as an established input because no feature list was supplied.

A reproducible linkage is nonetheless specifiable from public documentation. NCI's image information dictionary provides the participant identifier, screening year, study and series identifiers needed to map an image series to a participant and round, after which the participant record attaches [35]. The manifest that results must record which reconstruction was selected, admit only covariates available at prediction time, and exclude diagnostic workup, pathology and treatment fields, which may define labels but would leak outcomes if used as predictors. The protocol standardises continuous variables with training statistics and encodes categorical variables as one-hot vectors or embeddings, but does not state per-field encodings, missingness rates, imputation or the resulting clinical dimension. A concrete completion, namely training-median imputation with explicit missingness indicators for continuous fields, and a reserved missing category for categorical fields — is proposed in Section 11 as a rerun configuration, not asserted as historical procedure.

3.4 Image preprocessing and leakage controls

Volumes are resampled to 1 mm isotropic spacing with trilinear interpolation, clipped to $[-1000, 400]$ HU and mapped to $[0, 1]$; automatic lung masks restrict candidate extraction to the pulmonary fields; augmentation comprises three-dimensional flips, small rotations and mild intensity jitter.

Table 1: Methodological lineage. Each row states the prior idea, what HAMLET does with it, the internal experiment that tests the change, and what that experiment shows.

Prior idea	HAMLET use	Isolating experiment	What the experiment shows
Feature pyramids and query-based set prediction [5, 6]	Multi-scale differentiable lesion tokens with an objectness gate	Replace with hard ROI crops from a thresholded detector	Largest single AUROC change (-0.012). Does not isolate differentiability unless candidate count, encoder and crop context are matched; lesion-localisation IoU falls by 0.082 under the same replacement
Graph attention and learned discrete structure [7, 8]	Task-dependent lesion connectivity via relaxed Bernoulli edges	Replace with fixed k-NN topology ($k = 8$)	Second-largest change (-0.008); clearest evidence for learned connectivity, the learned graph settles near 25% density against a fixed eight-neighbour budget
Multimodal fusion [4, 27]	Clinical covariate node inside the lesion graph	Ablate the clinical node	The node conditions lesion processing before pooling; isolating its contribution is reported in Section 11 as ongoing work
Sparse expert routing [11, 12]	Top-two of four experts per lesion token	Single shared transformation	Small change (-0.002, unadjusted $p = 0.089$); routing histograms are non-uniform, consistent with representational diversity rather than proven phenotype specialisation
Adaptive computation time [13]	Representation-dependent halting, $r^* \leq 4$	Fixed four-round reasoning	Identical discrimination to three decimals at 2.25 versus 4.00 mean rounds; a resource-allocation effect, not an accuracy effect
Evidential classification [14, 15]	Dirichlet head with vacuity score	Softmax head with the same representation	-0.001 AUROC but $+0.042$ ECE; the cleanest single-mechanism attribution in the study

The lung-segmentation implementation, crop and padding policy, orientation convention and the distribution of slice depths are unspecified.

Three controls are described in our experiments and are necessary for the reported comparison to mean anything: all examinations and reconstructions from a participant stay in one partition; transform fitting and checkpoint selection use training and validation data only; and the retrieval index, when enabled, is built from training cases alone. Their execution cannot be verified without manifests. Two further conditions apply to the auxiliary data: detection evaluation must exclude patients seen during warm-up, and LUNA16 membership must be reconciled against any additional LIDC-IDRI training. Finally, a second NLST-derived collection is not independent external validation merely because it carries a different name — a point that governs Section 9.

Table 2: Dataset roles. Source-wide sizes describe the public resources, not the number of cases HAMLET processed.

Resource	Role in this study	Reference standard
NLST [1]	Primary multimodal training and held-out assessment; 2,068 reported test examinations	Release, eligible participants, executed join, endpoint horizon, split identifiers, per-partition counts
LUNA16 [22]	Image-only token-bank warm-up (50 epochs); intended candidate-sensitivity check	Exact training folds; a separate held-out detection evaluation
LIDC-IDRI [23]	Parent resource of LUNA16; stated auxiliary annotations	Whether any non-LUNA16 scans or ratings were used
Secondary cohort	Exploratory numerical summary only (Section 9)	Identity, independence, size, labels, prevalence, scanners

4 Method and mathematical formulation

4.1 Notation and scope

An examination comprises a volume $V \in \mathbb{R}^{H \times W \times D}$, a clinical vector $c \in \mathbb{R}^{d_c}$ and a binary label $y \in \{0, 1\}$. The model returns a cancer probability $\hat{p}$, a vacuity score u and a stopping round r_*. We write $C = 768$ for the feature width, $M = 4$ for the expert count, $J = 2$ for the number of classes and $R_{\max} = 4$ for the maximum depth. Separating M from J avoids a collision that arises when one symbol carries both, which makes the uncertainty expression read as though it depended on the expert count.

The formulation below preserves the documented architecture and makes its incomplete operations explicit. Equations marked as completions — the binary edge relaxation, the load-balancing objective, the cross-scale operator, the pooling rule and the differentiable halting objective — are mathematically coherent specifications of operations the source describes only in words. They must be reconciled with code before being attributed to the runs in Section 6; corrected mathematics is not evidence about what was executed.

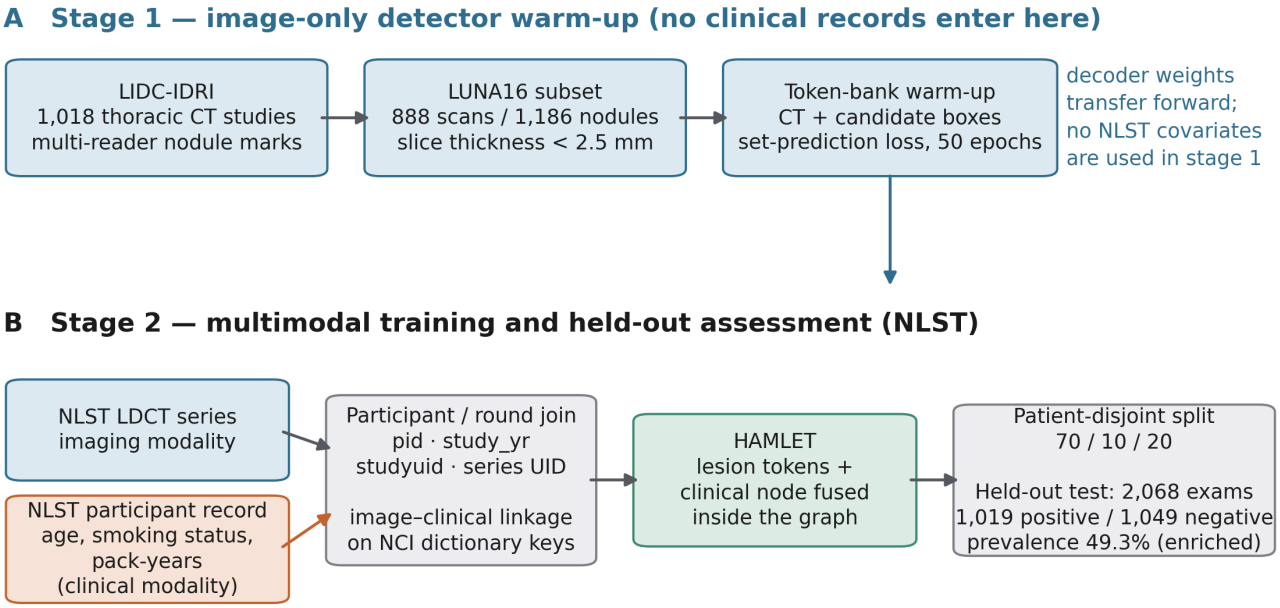


Figure 1: Dataset and modality flow. Stage 1 is image-only: LUNA16 candidate annotations supervise the token decoder and only weights cross into stage 2. Stage 2 is the multimodal stage, in which an NLST LDCT series and the participant's clinical record are joined and evaluated on a patient-disjoint split. Boxes reproduce the design stated in the source manuscripts; the join fields follow the NCI data dictionaries [33, 34, 35].

4.2 Image encoding and multi-resolution tokens

A volumetric encoder and feature pyramid [5] produce three scales,

$$F_s = \mathrm{FPN}_s(E_\theta(V)), \quad X_s = \mathrm{flatten}(F_s) \in \mathbb{R}^{L_s \times C}, \tag{1}$$

with $L_s = H_s W_s D_s$ for $s \in \{\text{fine}, \text{mid}, \text{coarse}\}$. E_θ is a 3-D convolutional backbone; the dense baseline instead uses Swin-UNETR [26], so encoder family is one of the axes on which that comparator differs.

Learned queries $Q_s \in \mathbb{R}^{N_s \times C}$ summarise spatial features through cross-attention. For a single head,

$$T_s = \mathrm{softmax}\left(\frac{(Q_s W_Q)(X_s W_K)^\top}{\sqrt{d_a}}\right) X_s W_V W_O, \tag{2}$$

with $W_Q, W_K, W_V \in \mathbb{R}^{C \times d_a}$, $W_O \in \mathbb{R}^{d_a \times C}$ and the softmax taken over the L_s spatial positions. Unlike a fixed crop, the resulting token retains a differentiable dependence on the attended feature map, which is precisely the property the hard-ROI ablation removes.

Token retention uses an objectness score and a soft gate. A specification compatible with the source description is

$$\begin{aligned} o_i &= \sigma(w_o^\top t_i + b_o), \quad q_i = \sigma\left(\frac{o_i - \eta}{\tau_t}\right), \\ h_i^{(0)} &= q_i t_i, \end{aligned} \tag{3}$$

where η is a selection threshold and $\tau_t > 0$ controls gate softness. The soft count $N_{\mathrm{eff}} = \sum_i q_i$ has an expected value near 64 in the reported configuration, corresponding to a 288:1 reduction from 18,432 feature locations. Two consequences follow and are easy to overstate. A deterministic soft sum is not a count of tensors physically removed, so any speed claim must be based on measured active tokens rather than on small gate weights; and if every image candidate is suppressed, pooling must remain defined, which requires a fallback token or reliance on the clinical node. In practice we retain the clinical node as the pooling fallback, so the pooled state stays defined even when every image candidate is suppressed.

During warm-up a candidate head maps each token to a centre and extent. A compatible set-prediction objective [6] is

$$\begin{aligned} \mathcal{L}_{\mathrm{det}} = \min_{\pi} \sum_{j=1}^{m} \Big[\ell_{\mathrm{obj}}(o_{\pi(j)}, 1) + \lambda_b \| b_{\pi(j)} - b_j^{\mathrm{gt}} \|_1 \Big] \\ + \sum_{i \notin \pi} \ell_{\mathrm{obj}}(o_i, 0), \end{aligned} \tag{4}$$

where π matches m reference nodules to queries. The source confirms set-prediction warm-up but not these exact terms, and Eq. (4) applies to initialisation only; no NLST lesion boxes are assumed to exist.

4.3 Clinical encoding and the fusion point

For continuous feature j, standardisation and the node encoder are

$$\begin{aligned} \tilde{c}_j &= \frac{c_j - \mu_j^{\mathrm{tr}}}{\max(s_j^{\mathrm{tr}}, \delta)}, \\ g^{(0)} &= \mathrm{LN}\{W_2\, \mathrm{GELU}(W_1 \tilde{c} + b_1) + b_2\}, \end{aligned} \tag{5}$$

with $\delta > 0$ guarding zero-variance features, $W_1 \in \mathbb{R}^{d_g \times d_c}$ and $W_2 \in \mathbb{R}^{C \times d_g}$ mapping the clinical vector into the token feature space. The node is appended as $h_0^{(0)} = g^{(0)}$ and connected to the lesion nodes. This is the fusion point: messages into a

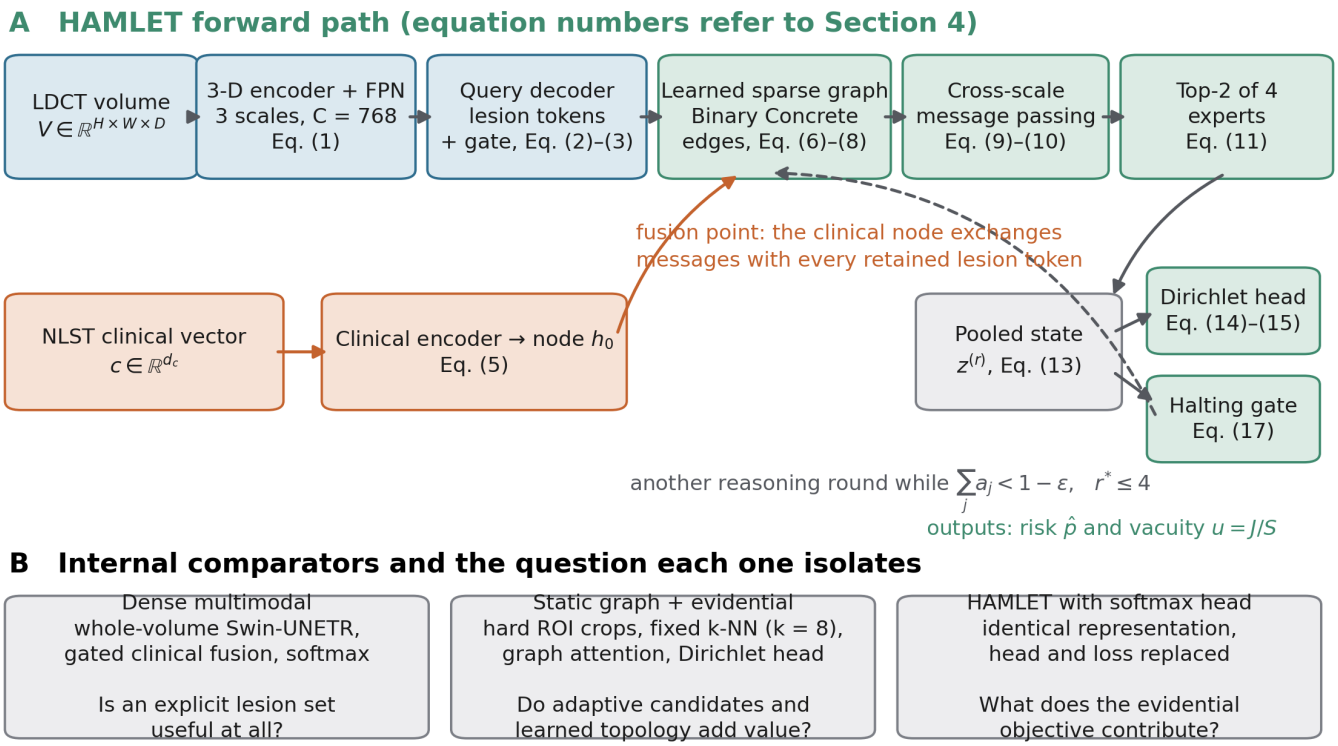


Figure 2: HAMLET and its internal comparators. (A) The forward path, annotated with the equations of Section 4. Clinical covariates enter as a graph node, so fusion precedes pooling and prediction; the evidential head and the halting gate read the same pooled state but produce different outputs, which is why halting is representation-dependent rather than uncertainty-driven. (B) The three internal comparators and the question each isolates. The diagram is a method schematic, not an execution trace.

lesion representation can depend on clinical covariates, and the pooled graph state determines the prediction. No patient correspondence between LUNA16 and NLST is required or would be appropriate.

The clinical node has no centroid, so the geometric term in Eq. (6) applies to lesion–lesion pairs only, and clinical edges need a separate rule. We define them as bidirectional and geometry-free, connecting the clinical node to every retained lesion token.

4.4 Learned lesion connectivity

For distinct lesion nodes i, j with centres $p_i, p_j \in \mathbb{R}^3$, the affinity is

$$\phi_{ij} = f_\phi\left([W_p t_i \,\|\, W_p t_j \,\|\, (p_i - p_j)]\right), \quad W_p \in \mathbb{R}^{d_p \times C}, \tag{6}$$

with $f_\phi : \mathbb{R}^{2d_p+3} \to \mathbb{R}$ producing edge log-odds. Ordered concatenation permits $\phi_{ij} \neq \phi_{ji}$, so the scorer is directed; whether the implementation later symmetrises the adjacency or adds self-loops is a detail we fix explicitly: the adjacency is used as scored, without symmetrisation, and self-loops are not sampled separately.

For a Bernoulli edge, the standard Binary Concrete relaxation [10, 9] is

$$\begin{aligned} \ell_{ij} &= \log U_{ij} - \log(1 - U_{ij}), \\ \tilde{A}_{ij} &= \sigma\left(\frac{\phi_{ij} + \ell_{ij}}{\tau_g}\right), \end{aligned} \tag{7}$$

with $U_{ij} \sim \mathrm{Uniform}(0,1)$. This is a correction. Our experiments add a single Gumbel variate inside the sigmoid, which is not the binary relaxation: logistic noise is the *difference* of two independent Gumbels, so the single-Gumbel expression induces a shifted, asymmetric edge distribution. If the historical code used that expression, the behaviour must be described accurately and the corrected sampler evaluated in a rerun. The temperature is annealed from 2.0 to 0.1; whether edges are sampled, thresholded or replaced by deterministic probabilities at inference is unrecorded.

A sparsity regulariser keeps the graph compact,

$$\Omega_{\text{edge}} = \sum_{i \neq j} \tilde{A}_{ij}, \qquad \mathscr{L}_{\text{sparse}} = \lambda_{\text{sparse}} \Omega_{\text{edge}}, \tag{8}$$

with the coefficient appearing exactly once. An earlier formulation included λ_{sparse} both inside the penalty and again in the total objective, which would double-weight it. Normalising by the number of candidate edges is a defensible alternative when token counts vary, but it rescales the coefficient and is not substituted silently. The reported 25% density also needs a definition: mean soft weight and fraction of non-zero hard edges are different quantities.

4.5 Message passing and cross-scale exchange

Graph attention over the allowed neighbour set gives

$$\begin{aligned} a_{ij}^{(r)} &= \frac{\exp s_\theta\left(h_i^{(r)}, h_j^{(r)}\right)}{\sum_{k \in \mathcal{N}(i)} \exp s_\theta\left(h_i^{(r)}, h_k^{(r)}\right)}, \\ m_i^{(r)} &= \sum_{j \in \mathcal{N}(i)} \tilde{A}_{ij}\, a_{ij}^{(r)}\, W_v h_j^{(r)}, \end{aligned} \tag{9}$$

retaining the multiplicative adjacency–attention form; a normalised edge-weighted softmax would be a different operator and is not assumed. For the fixed-topology ablation, $\tilde{A}$ is replaced by a fixed nearest-neighbour mask while every other operator is held constant.

The cross-scale operator, left undefined in our experiments, admits the dimensionally explicit completion

$$\begin{aligned} C_i^{(r)} &= \sum_{s \neq s(i)} \sum_{j:s(j)=s} \beta_{ij}^{(r)} W_s h_j^{(r)}, \\ h_i^{(r+1)} &= h_i^{(r)} + m_i^{(r)} + \gamma C_i^{(r)}, \end{aligned} \tag{10}$$

where β_{ij} normalises over eligible tokens at the source scale, $W_s \in \mathbb{R}^{C \times C}$ projects the message and γ controls cross-scale strength. Coarse-to-fine context and fine-to-coarse aggregation alternate across rounds. Removing this path tests whether resolution-specific context contributes anything beyond ordinary graph attention.

4.6 Expert routing

For a lesion state h_i,

$$\begin{aligned} r_i &= \mathrm{softmax}(W_r h_i + b_r) \in \mathbb{R}^M, \\ \hat{h}_i &= \sum_{k \in \mathrm{Top2}(r_i)} r_{ik} E_k(h_i), \end{aligned} \tag{11}$$

with $E_k : \mathbb{R}^C \to \mathbb{R}^C$ and $M = 4$. Eq. (11) preserves the unrenormalised selected weights; renormalising over the two selected experts would change the output scale and needs confirmation. Top-two execution halves expert evaluations relative to running four otherwise identical experts — not relative to a single shared transformation, and not relative to the whole model.

A concrete load-balancing completion in the style of [12] is

$$\begin{aligned} f_k &= \frac{1}{2B_t} \sum_{i=1}^{B_t} \mathbf{1}\{k \in \mathrm{Top2}(r_i)\}, \quad \bar{r}_k = \frac{1}{B_t} \sum_i r_{ik}, \\ \mathscr{L}_{\mathrm{bal}} &= M \sum_k f_k \bar{r}_k, \end{aligned} \tag{12}$$

where B_t is the number of routed tokens in the batch. Balanced utilisation is a training property; it says nothing about whether the resulting partition is clinically meaningful.

4.7 Pooling and evidential prediction

Pooling is unspecified in the source. One compatible completion concatenates gate-weighted lesion pooling with the updated clinical node,

$$\begin{aligned} v^{(r)} &= \frac{\sum_i q_i \hat{h}_i^{(r)}}{\sum_i q_i + \delta}, \\ z^{(r)} &= P\Big(\Big[v^{(r)} \,\|\, h_0^{(r)} \Big] \Big) \in \mathbb{R}^C, \end{aligned} \tag{13}$$

with trainable $P : \mathbb{R}^{2C} \to \mathbb{R}^C$. The evidential head is then

$$\begin{aligned} e^{(r)} &= \mathrm{softplus}(W_e z^{(r)} + b_e), \\ \alpha_k^{(r)} &= 1 + e_k^{(r)}, \qquad S^{(r)} = \sum_{k=1}^{J} \alpha_k^{(r)}, \end{aligned} \tag{14}$$

and prediction and vacuity follow as

$$\hat{p}_k^{(r)} = \alpha_k^{(r)} / S^{(r)}, \qquad u^{(r)} = J / S^{(r)}. \tag{15}$$

The Dirichlet marginal variance,

$$\mathrm{Var}(p_k \mid \alpha) = \frac{\alpha_k (S - \alpha_k)}{S^2 (S+1)}, \tag{16}$$

is a distributional quantity, not an observed error probability. We use *vacuity* rather than *epistemic uncertainty* for J/S deliberately: the score earns an epistemic interpretation only to the extent that held-out error detection, calibration and shift experiments support it, and Section 8 reports the held-out evidence that bears on that interpretation together.

4.8 Adaptive depth

The documented halting rule is

$$\begin{aligned} a_r &= \sigma(w_h^\top z^{(r)} + b_h), \\ r_* &= \min\left(\left\{ r : \sum_{j \le r} a_j \ge 1 - \varepsilon \right\} \cup \{R_{\max}\} \right), \end{aligned} \tag{17}$$

the union forcing a stop at maximum depth when the threshold is never reached. Because $u^{(r)}$ does not appear in Eq. (17), the description "uncertainty-driven halting" would not be accurate; the gate and the evidential head merely share an input. The value of ε, selected on the validation partition and then held fixed, is reported with the remaining hyperparameters in Supplementary Section S2.

A hard integer r_* supplies no ordinary gradient to the gate. An ACT-style completion [13] assigns $\pi_r = a_r$ for $r < r_*$, $\pi_{r_*} = 1 - \sum_{j<r_*} a_j$ and zero mass thereafter, giving

$$\begin{aligned} \mathscr{L}_{\mathrm{pred}} &= \sum_{r=1}^{r_*} \pi_r \mathscr{L}_{\mathrm{edl}}^{(r)}, \\ \mathscr{L}_{\mathrm{ponder}} &= \sum_{r=1}^{r_*} r\, \pi_r. \end{aligned} \tag{18}$$

Away from changes in the stopping boundary this is differentiable through the gate weights. It is one proposal, not a restatement of what was trained; terminal-head inference and round-mixture inference are different designs, and any change between them requires re-evaluation.

4.9 Objective

The evidential term is the expected cross-entropy under the Dirichlet [14],

$$\mathscr{L}_{\mathrm{edl}} = -\mathbb{E}_{p \sim \mathrm{Dir}(\alpha)} \sum_k y_k \log p_k = \sum_k y_k \left[\psi(S) - \psi(\alpha_k) \right], \tag{19}$$

with ψ the digamma function and y one-hot. This is not the squared-error-plus-variance variant that also travels under the name evidential Bayes risk, and the distinction changes the gradient. Writing $\tilde{\alpha}_k = y_k + (1 - y_k)\alpha_k$ and $\tilde{S} = \sum_k \tilde{\alpha}_k$, the vacuity regulariser is

$$\begin{aligned} \mathscr{L}_{\mathrm{KL}} = \log\Gamma(\tilde{S}) - \sum_k \log\Gamma(\tilde{\alpha}_k) - \log\Gamma(J) \\ + \sum_k (\tilde{\alpha}_k - 1) \left[\psi(\tilde{\alpha}_k) - \psi(\tilde{S}) \right], \end{aligned} \tag{20}$$

which is $\mathrm{KL}[\mathrm{Dir}(\tilde{\alpha}) \,\|\, \mathrm{Dir}(\mathbf{1})]$. Its coefficient is annealed linearly; neither the final value nor the schedule duration is recorded. The total objective, in notation that avoids double-counting sparsity, is

$$\begin{aligned}\mathscr{L} = \mathscr{L}_{\text{pred}} + \lambda_{\text{KL}}(t)\mathscr{L}_{\text{KL}} + \lambda_{\text{sparse}}\Omega_{\text{edge}} \\ + \lambda_{\text{bal}}\mathscr{L}_{\text{bal}} + \lambda_{\text{ponder}}\mathscr{L}_{\text{ponder}}.\end{aligned} \tag{21}$$

The original formulation used a terminal evidential loss with a hard r_*; Eqs. (18) and (21) show how a trainable version would be specified. No numerical coefficient is invented, and the warm-up objective of Eq. (4) belongs to initialisation, not to the NLST stage.

4.10 Optional retrieval prior

An optional training-only index [28] contributes evidence from nearby training representations. To avoid double-counting the unit prior, an explicit evidence formulation is

$$\begin{aligned}\alpha^* = \mathbf{1} + e^{(r_*)} + \kappa \sum_{j \in \mathscr{R}} w_j \,\mathrm{onehot}(y_j), \\ w_j \geq 0, \quad \sum_j w_j = 1,\end{aligned} \tag{22}$$

with neighbour count, distance weighting and evidence scale κ unspecified. Retrieval neighbours must exclude validation and test cases, and during training the query itself.

A configuration ambiguity follows from this module and is not cosmetic. The ablation table labels a lower-performing configuration "without retrieval" while the prose describes retrieval as non-default. Until run configurations resolve it, we call 0.935 the *reported full configuration* and 0.933 the *reported retrieval-removed configuration*, and we do not silently reassign the better number to a retrieval-free system.

5 Experimental protocol and baseline hierarchy

5.1 Internally trained comparators

The internal comparison has three levels, sketched in Fig. 2B and defined in Table 3: a dense multimodal reference, a static graph with evidential prediction, and matched component replacements inside HAMLET. Published systems form a separate contextual group and are never described as baseline reruns. The dense baseline is not a reproduced M3FM, and the static graph is an internal reference rather than a named published method.

We use common patient partitions, preprocessing, augmentation, AdamW optimisation [29] with a cosine schedule after five warm-up epochs, peak learning rate 3×10^{-4}, minimum 10^{-6}, mixed precision, and checkpoint selection on validation AUROC with patience 20. Training used an NVIDIA A100 80 GB, and HAMLET required 417 GPU-hours. The complete hyperparameter set is given in Supplementary Section S2.

Shared settings are necessary for fairness but not sufficient. A 312 M-parameter dense comparator against a 104 M-parameter HAMLET does not isolate graph structure, and moving from hard ROIs to learned queries simultaneously changes candidate recall, spatial context and pre-training exposure. The strongest additional control, a matched image backbone with identical clinical inputs and prediction head, varying only tokenisation and topology — supplements rather than replaces the existing baselines, and is specified in Section 11.

5.2 Metrics and statistical units

AUROC measures ranking across thresholds and is prevalence-insensitive. AUPRC is not: it depends directly on the 49.3% positive rate of this cohort, so its value here is not transferable to a screening population. We report AUPRC as trapezoidal area, and the reported values are retained under their original name until that is recovered.

Threshold metrics must come from one prediction vector at one threshold. For counts TP, TN, FP, FN the audit uses

$$\begin{aligned}\mathrm{Acc} &= \frac{TP+TN}{n}, & \mathrm{PPV} &= \frac{TP}{TP+FP}, \\ \mathrm{Sens} &= \frac{TP}{TP+FN}, & \mathrm{Spec} &= \frac{TN}{TN+FP}, \\ F_1 &= \frac{2TP}{2TP+FP+FN}.\end{aligned} \tag{23}$$

Sensitivity and recall are the same quantity and are not counted as separate achievements.

Calibration is reported as expected calibration error over ten equal-width bins,

$$\mathrm{ECE} = \sum_{b=1}^{10} \frac{|B_b|}{n} \left| \frac{1}{|B_b|} \sum_{i \in B_b} a_i - \frac{1}{|B_b|} \sum_{i \in B_b} c_i \right|, \tag{24}$$

where for top-label calibration $c_i = \max_k \hat{p}_{ik}$ and a_i is correctness, while for binary risk calibration $c_i = \hat{p}_i$ and $a_i = y_i$. These are different estimands, and the source does not say which was computed. The gap matters because the two disagree systematically at balanced prevalence. A complete calibration report additionally needs

$$\begin{aligned}\mathrm{Brier} &= \frac{1}{n}\sum_i (\hat{p}_i - y_i)^2, \\ \mathrm{NLL} &= -\frac{1}{n}\sum_i \left[y_i \log \hat{p}_i + (1-y_i)\log(1-\hat{p}_i)\right],\end{aligned} \tag{25}$$

together with adaptive-bin estimates and bin counts [19, 18]; none is available.

Confidence intervals come from 2,000 outcome-stratified bootstrap resamples at examination level, paired AUROC comparisons from DeLong's test [30] and paired classification changes from McNemar's test [31]. Two cautions apply. If participants contribute repeated examinations, examination-level resampling understates dependence and the bootstrap should cluster by patient; and bootstrap variability for one fitted model is not variability across training seeds, so no seed-level dispersion can be recovered from the reported intervals. A valid McNemar test also needs the two models' discordant predictions on the same cases, which marginal error counts do not provide. The p-values are therefore reported as unadjusted descriptive statistics. A final analysis should prespecify primary comparisons and control the remaining family, for instance with Holm's procedure [32].

Table 3: Internal baseline hierarchy. Each comparator is defined by what it holds fixed relative to HAMLET, which determines the question it can answer.

Model	Inputs and architecture	Objective	Question isolated	Planned control
Dense multimodal	Whole-volume CT through a 3-D Swin-UNETR encoder [26]; clinical embeddings fused by gated multimodal units [27]	Softmax head; cross-entropy	Is an explicit lesion set useful relative to volumetric fusion?	Backbone configuration, pre-training, exact loss; 312 M versus 104 M parameters is not a matched comparison
Static graph + evidential	Hard-thresholded ROI crops; fixed k-NN graph, $k = 8$; graph attention; Dirichlet head	Evidential loss family	Do adaptive candidates and learned topology add value once a relational model and evidential head exist?	Clinical entry point, depth, auxiliary losses, candidate-matching rule
Reported full HAMLET	Multi-scale tokens, clinical node, learned graph, cross-scale exchange, top-two experts, adaptive depth, evidential head	Eq. (21)	Combined architecture	Retrieval status unresolved (Section 4)
HAMLET + softmax	Identical representation; evidential head and loss replaced	Softmax classification	What does the evidential objective contribute?	No temperature-scaled softmax control [18]
Component replacements	One documented component removed or replaced	Remaining setup retained	Mechanism-specific attribution	Seeds, capacity matching, configuration diffs

6 Results

6.1 Discrimination and calibration

Table 4 and Fig. 3 give the held-out comparison. HAMLET exceeds the dense encoder by 0.021 AUROC and 0.026 AUPRC, and the static graph by 0.012 AUROC and 0.014 AUPRC, with paired DeLong tests giving $p = 0.003$ and $p = 0.018$ respectively. The second margin is the informative one. The dense encoder differs from HAMLET in representation, capacity and predictive head at once, whereas the static graph already has candidate-based relational reasoning and a Dirichlet output, so the 0.012 gap is the part attributable to adaptive candidate selection and learned connectivity.

The ECE differences are 0.044 against the dense baseline and 0.010 against the static graph, relative reductions of 53.0% and 20.4%. The dense-baseline contrast is the less informative of the two, because that model differs in representation and capacity as well as in its head. The softmax ablation of Section 7, which changes only the head and the loss, is the comparison that actually isolates evidential prediction.

6.2 Operating point

At the validation-selected threshold of 0.52, HAMLET produces $TP = 947$, $TN = 931$, $FP = 118$ and $FN = 72$ on the 2,068 held-out examinations, giving accuracy 90.81%, precision 88.92%, sensitivity 92.93%, specificity 88.75% and $F_1 = 0.9088$. Table 5 and Fig. 4 give the same decomposition for all three models. The error profile moves in the direction the area summaries predict: relative to the dense encoder, HAMLET converts 24 false negatives and 11 false positives, and relative to the static graph, 12 and 3.

One caveat attaches to this table. Each model sits at its own validation-selected threshold, so the comparison mixes ranking quality with threshold placement. Comparison at matched validation sensitivity would separate the two, and the threshold-free evidence in Table 4 is the stronger basis for the architectural claim.

6.3 Literature context

Table 6 separates published results from the internal experiments. Sharing the name NLST does not establish identical participants, labels, prevalence, preprocessing or follow-up, and the differences are large enough to make a naive ranking meaningless. Recent replication work notes, for example, that M3FM's lung-cancer risk evaluation used a non-standard NLST split, exactly the kind of protocol difference a leaderboard column would hide [25].

Two published AUROCs in Table 6 exceed HAMLET's. That does not establish their superiority on this task either, for the same reason in reverse: the experiments differ in cohort, endpoint, input availability and prevalence. A scientifically useful comparison retrains or re-evaluates compatible models on one patient-disjoint split with a common endpoint and matched covariate availability, and audits whether public pre-trained weights have already seen the proposed test participants. Until then, HAMLET's position relative to published risk models remains an open question, and Section 11 makes it a priority rather than a footnote.

7 Ablation and mechanism-specific interpretation

Table 7 reproduces the component replacements. The purpose of this section is not to restate the numbers but to state which

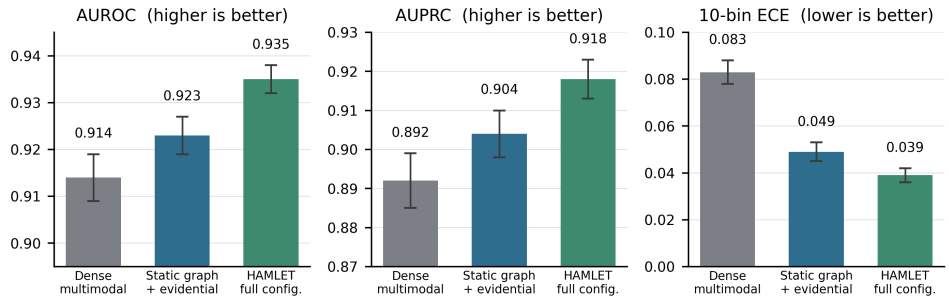


Figure 3: Primary held-out comparison on the outcome-enriched NLST test cohort ($n = 2{,}068$). Bars give point estimates with 95% bootstrap margins over 2,000 outcome-stratified resamples. HAMLET leads on all three summaries, but the margin over the static graph is far smaller than the margin over the dense encoder, which is the comparison that matters for the architectural claim.

Table 4: Primary results on the enriched NLST test cohort ($n = 2{,}068$; 49.3% positive). Parentheses give 95% bootstrap intervals. ΔAUROC is measured against HAMLET; DeLong p-values are unadjusted and should be read as descriptive, since no multiplicity correction is applied across the comparison family.

Model	AUROC	AUPRC	ECE (10-bin)	ΔAUROC	DeLong p
Dense multimodal	0.914 (0.909–0.919)	0.892 (0.885–0.899)	0.083 (0.078–0.088)	+0.021	$p = 0.003$
Static graph + evidential	0.923 (0.919–0.927)	0.904 (0.898–0.910)	0.049 (0.045–0.053)	+0.012	$p = 0.018$
HAMLET	**0.935** (0.932–0.938)	**0.918** (0.913–0.923)	**0.039** (0.036–0.042)	—	—

scientific question each row answers and what the answer is worth.

Token representation. Eqs. (2)–(3) allow the examination objective to shape candidate features and their gating. Replacing the bank with hard ROI crops produces the largest reported decrease, 0.012 AUROC and 0.017 AUPRC, which is consistent with candidate representation being the dominant factor. It does not isolate differentiability on its own: candidate count, encoder, warm-up exposure and crop context all change together, and a controlled version of this experiment would hold the first three fixed. Lesion-localisation IoU falls by 0.082 under the same replacement, which points the same way, though IoU measured against warm-up annotations is a weaker endpoint than the examination-level objective and we treat it as supporting rather than primary evidence.

Learned topology. Replacing Eq. (7) with a fixed graph costs 0.008 AUROC and 0.011 AUPRC, the clearest evidence in the record for task-dependent connectivity. One confound deserves emphasis because it is easy to miss: a fixed eight-neighbour graph and a learned graph at roughly 25% density need not carry the same edge budget, so part of the gap may reflect communication capacity rather than topology quality. A density-matched fixed graph is the required control.

Cross-scale exchange. Removing Eq. (10) costs 0.004 AUROC and 0.006 AUPRC, materially smaller than the two representation effects. We attribute the loss particularly to part-solid and ground-glass findings, which is consistent with the intended role of the coarse path, in which local texture identifies the candidate while the coarse token supplies segmental context — though we report the aggregate change rather than subgroup effects, and treat the phenotype-specific reading as a hypothesis for the factorial experiment described in Section 11.

Expert routing. Replacing the expert pool with a shared transformation costs 0.002 AUROC at an unadjusted $p = 0.089$, which does not support a conventional significance claim, and the parameter-count difference between the two configurations complicates even that. Routing histograms are non-uniform after load balancing, and different experts receive more solid, part-solid, ground-glass or calcified examples. Those groups were assigned by post-hoc inspection rather than by an independent test against phenotype labels, so this is a routing pattern rather than demonstrated specialisation. We have not swept the expert count, and make no claim that performance saturates at four.

Evidential prediction. Softmax replacement changes AUROC by 0.001 and raises ECE by 0.042. This contrast is the cleanest mechanism attribution in the study: the evidential formulation buys calibration, not ranking. Two qualifications keep it from being overstated. The ablation changes the loss as well as the head, so it measures the objective rather than the activation alone; and ECE by itself cannot establish better probability estimation. A validation-calibrated softmax comparator with Brier score and NLL, per Eq. (25), is the necessary next comparison, since temperature scaling recovers much of a softmax model's calibration at negligible cost [18].

Adaptive depth. Fixed-depth and adaptive models are identical in AUROC and AUPRC to three decimals while mean depth falls from 4.00 to 2.25, a 43.75% reduction in reasoning rounds. Rounded equality is not a non-inferiority test, and a pre-specified margin would be needed to make that claim. The timing contrast of 134 versus 104 ms is a 22.4% reduction relative to fixed depth, or a 28.8% increase in the

Table 5: Held-out operating point at the validation-selected threshold ($n = 2{,}068$; 1,019 positive, 1,049 negative). Precision and F_1 are reported at the enriched prevalence of this cohort and do not transfer to screening conditions; Section 8 quantifies that gap.

Model	TP / TN / FP / FN	Acc.%	Prec.%	Sens.%	Spec.%	F_1
Dense multimodal	923 / 920 / 129 / 96	89.12	87.74	90.58	87.70	0.8914
Static graph + evidential	935 / 928 / 121 / 84	90.09	88.54	91.76	88.47	0.9012
HAMLET	947 / 931 / 118 / 72	90.81	88.92	92.93	88.75	0.9088

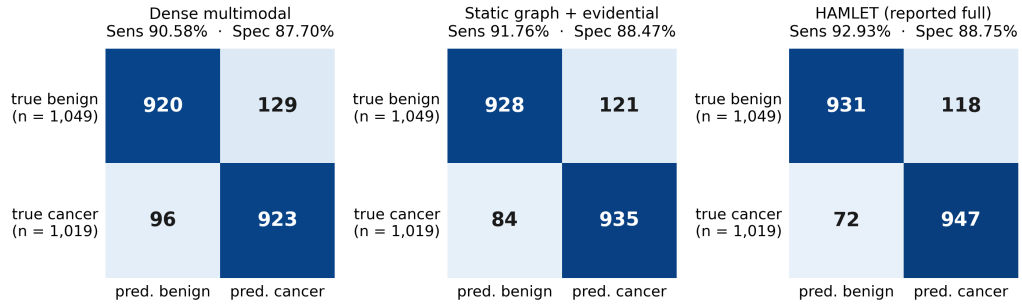


Figure 4: Held-out confusion matrices at the operating threshold. The gain concentrates in the false-negative cell, which is the clinically consequential direction in a screening task.

other direction. The direction quoted depends on which configuration is the reference, so we state both.

Retrieval. The table implies a gain of 0.002 AUROC and 0.003 AUPRC at a timing increase from 88 to 104 ms, 18.2% relative to the retrieval-removed model. A small discrimination change bought with extra latency is a natural trade-off and an argument against enabling retrieval by default. Retrieval is therefore not enabled in the default configuration, and a frozen retrieval-on/off evaluation should precede any deployment recommendation.

Single-removal effects are conditional on the rest of the architecture and are not additive. Their sum, 0.029 AUROC, exceeds the gain over either baseline (0.012 and 0.021). That is not evidence of an arithmetic error; it is what one expects from interacting components, but it does rule out reading the rows as independent shares of a total improvement, and Fig. S2 in the supplement makes the point visually. A 2×2 factorial over tokenisation and topology would separate the two dominant mechanisms properly.

8 Calibration, uncertainty and prevalence transfer

The reported ECE improvement is encouraging within the cohort that produced it, but screening involves a different class prior and possibly a different case spectrum. Under the restrictive assumption that only class priors change, a calibrated source probability transfers by a logit shift,

$$\operatorname{logit} p_t = \operatorname{logit} p_s + \operatorname{logit} \pi_t - \operatorname{logit} \pi_s, \tag{26}$$

with π_s and π_t the source and target prevalences. Eq. (26) is an identity, not a result produced by HAMLET, and enrichment based on lesion severity, site or follow-up can violate its assumption outright.

The same caution applies with more force to predictive value. If sensitivity and specificity were transferable, $\mathrm{PPV}(\pi) = \mathrm{Sens}\,\pi / [\mathrm{Sens}\,\pi + (1 - \mathrm{Spec})(1 - \pi)]$. Using the count-derived HAMLET operating point at an illustrative 1% prevalence gives a PPV near 7.7%, against 88.9% in the enriched table, an order-of-magnitude collapse visible in Fig. 6. The high apparent precision of a balanced evaluation says nothing about clinical utility, and this is the single most important reason the enriched cohort must be described as such wherever its numbers appear.

Vacuity separates correct from incorrect predictions in the expected direction: mean u is 0.11 over correctly classified examinations and 0.47 over misclassified ones. Deferring the most uncertain 20% of cases raises accuracy on the retained subset, which is the behaviour a selective-prediction rule requires. Two qualifications bound how far that observation travels. A mean is a weak summary of a score whose usefulness depends on its whole distribution, and the separation is measured at 49.3% prevalence, where both the error mix and the value of deferral differ from screening conditions. Supplementary Section S3 reports the depth and vacuity distributions behind these means.

A complete reanalysis should report Brier score and NLL as in Eq. (25), reliability diagrams with bin counts, both fixed- and adaptive-bin calibration, and a calibration slope and intercept fitted on a held-out calibration sample, which separates overconfidence from ranking quality. For a deferral rule retaining cases with $u_i \le t$, coverage and retained risk are

$$\begin{aligned} C(t) &= \frac{1}{n} \sum_i \mathbf{1}(u_i \le t), \\ R(t) &= \frac{\sum_i \mathbf{1}(u_i \le t)\mathbf{1}(\hat{y}_i \ne y_i)}{\sum_i \mathbf{1}(u_i \le t)}, \end{aligned} \tag{27}$$

and a risk–coverage curve with patient-clustered bootstrap intervals [20] is far more informative than a single mean score. A triage rule must also define whether "missed cancer" is divided by all cancers, retained cancers or all retained exami-

Table 6: Literature context. AUROCs are paper-reported values under each study's own protocol. Omission of AUPRC or ECE means a matched endpoint-specific value was not verified, not that the study never reported one. The right-hand column states why a direct numerical ranking would be invalid.

Study	Dataset and task	Inputs and method	Reported result	Relationship to HAMLET
Ardila et al. [2]	6,716 NLST cases; cancer prediction with reader comparison	Current CT and available prior CT; 3-D CNN	AUROC 0.944 (NLST)	Strong image-only reference. Uses prior imaging; endpoint and cohort must be harmonised before comparison
Sybil [3]	NLST, MGH, CGMH; one-year and longer-horizon risk	Single LDCT, no clinical inputs	One-year AUROC 0.92 / 0.86 / 0.94	Longitudinal risk endpoint; HAMLET's horizon is undefined, so ranking is not possible
M3FM [4]	NLST-derived risk task within a 17-task study	Multiscale CT plus 49 clinical data types	One-year AUROC 0.9400 (0.9119–0.9698)	Closest multimodal lineage; evaluated on a non-standard split [25]; our dense comparator is not M3FM
DeepSEED [21]	LUNA16 / LIDC-IDRI; nodule detection	3-D proposal detector with channel recalibration	FROC-based detection evaluation	Candidate-generation reference for the warm-up stage; detection sensitivity is not examination AUROC
Tushar et al. [24]	DLCS training; nodule classification across four corpora	Detection-informed ResNet50 warm-start	AUROC 0.71 (DLCS), 0.90 (LUNA16), 0.81 (NLST-3D), 0.80 (LUNA25)	Quantifies how strongly this metric depends on the evaluation corpus; lesion- and examination-level endpoints differ
HAMLET (this record)	Enriched NLST test set, 2,068 examinations	CT plus clinical node; adaptive lesion graph	AUROC 0.935; AUPRC 0.918; ECE 0.039	Internal comparison; protocol-matched evaluation pending

Table 7: component replacements, relative to the reported full configuration. p-values are unadjusted and descriptive. Latency is model-only forward time.

Configuration	AUROC	AUPRC	ECE	ms
HAMLET	**0.935**	**0.918**	**0.039**	104
Hard ROI replacement	0.923 (p <0.001)	0.901	0.042	91
Fixed k-NN topology	0.927 (p =0.004)	0.907	0.042	98
No cross-scale path	0.931 (p =0.021)	0.912	0.041	101
Shared head (no MoE)	0.933 (p =0.089)	0.914	0.040	100
Softmax head	0.934 (p =0.18)	0.916	0.081	102
Fixed four rounds	0.935 (p =0.11)	0.918	0.039	134
Retrieval removed	0.933 (p =0.076)	0.915	0.040	88

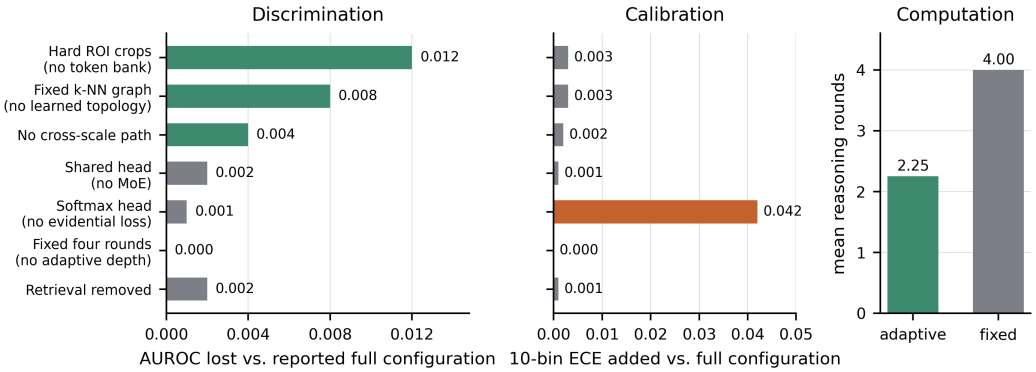
nations. The summary of 58% automation at a 1.3% missed-cancer rate lacks that denominator and merges automatic with expedited review, so it is excluded from this manuscript's claims entirely rather than repeated with a caveat.

9 Secondary cohort and distribution shift

Our experiments preserve a secondary-cohort summary: AUROC 0.915, AUPRC 0.897, ECE 0.045 and accuracy 88.4%, alongside a scanner-subgroup AUROC spread of 1.3 percentage points and an ECE increase of 0.009 under moderate shift. The cohort's identity, size, collection period, outcome horizon, clinical schema and independence are nowhere recorded. These numbers are retained in the supplementary audit for traceability and are excluded from the main comparison, because a same-source subset, a scanner subgroup and an independent institutional cohort answer three different questions and only the third supports a generalisation claim.

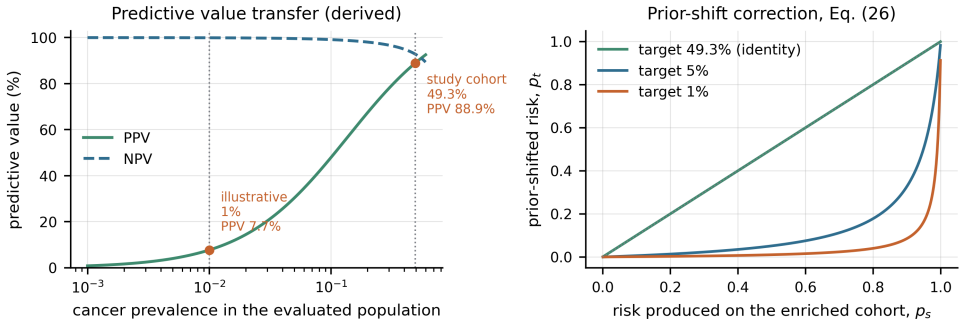
The corruption experiments are a different kind of evidence and deserve a more precise description than they received. Gaussian noise and motion blur test sensitivity to image degradation; they do not stand in for a new clinical population, and their parameters require a defined intensity scale, voxel units and application stage to be reproducible. The directionally encouraging observation, that vacuity rose as image quality fell, is the behaviour one wants from an uncertainty estimate, but it was measured on synthetic corruption of the same cohort and is reported here as such.

A defensible external validation freezes weights, preprocessing, calibration, thresholds and retrieval index before outcomes are accessed, then reports matched input availability, missingness, participant and examination counts, event prevalence, acquisition characteristics, label horizon and confidence intervals. DLCS is a reasonable candidate resource for lesion-level evaluation subject to task compatibility [24], but it cannot be assigned retrospectively as the unnamed cohort. Any NLST-derived collection, including NLST-3D or LUNA25, requires an explicit participant-overlap check: a different name does not establish independence from NLST training.



Each bar replaces one component and holds the rest of the architecture fixed. Single-removal effects are conditional on the remaining architecture and do not sum to the total margin over either baseline.

Figure 5: Component replacement separates three distinct roles. Representation-level components dominate discrimination (left), the evidential objective dominates calibration (centre), and adaptive halting acts only on computation (right). Differences are computed from the ablation table; no new uncertainty estimates are introduced.



Both panels are analytic consequences of the held-out operating point and of Eq. (26); neither is a measured screening result. They hold only if class-conditional feature distributions are stable across the two populations.

Figure 6: Why an enriched cohort cannot speak for a screening population. (Left) Positive and negative predictive value implied by the count-derived operating point across prevalence. (Right) The prior-shift correction of Eq. (26) applied to risks produced on the enriched cohort. Both panels are analytic consequences of the reported counts, not measured screening results, and hold only under stable class-conditional feature distributions.

Table 8: Computational profile. All measurements on a single A100 80 GB; latency is model-only forward time at batch size one and excludes DICOM loading, resampling and lung segmentation.

Quantity	Dense	Static graph	HAMLET
Parameters (M)	312	87	104
Peak memory (GB)	11.2	5.1	5.3
Model latency (ms/exam)	78	92	104
Mean reasoning rounds	n/a	4.00	2.25
Training (GPU-hours)	not reported	304	417

10 Computational analysis

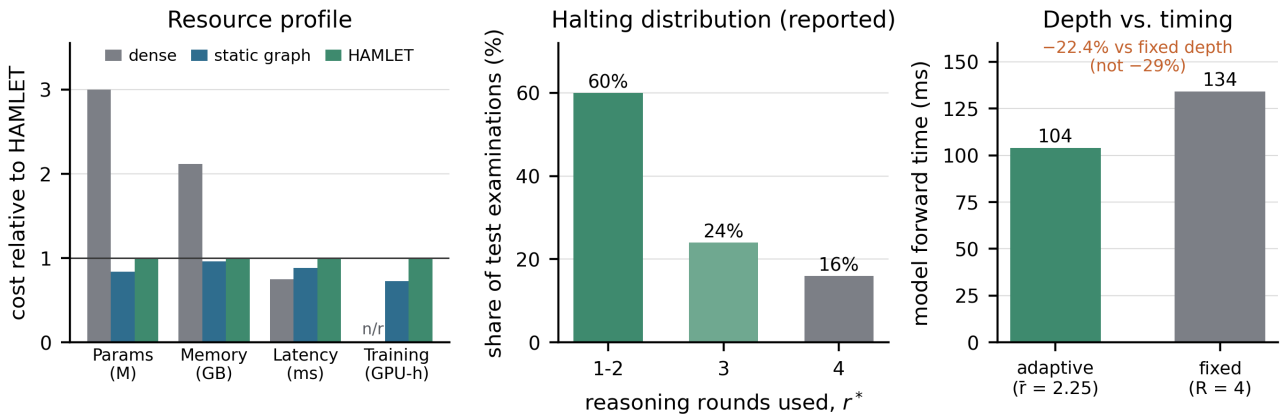
Table 8 collects these measurements. Relative to the dense baseline, HAMLET records 66.7% fewer parameters and 52.7% lower peak memory but 33.3% higher latency. Relative to the static graph it has 19.5% more parameters, 3.9% greater peak memory, 13.0% higher latency and 37.2% more training GPU-hours. The existing record therefore does not support a blanket efficiency claim, and this manuscript does not make one. What it supports is a parameter–memory trade-off against dense encoding, plus a within-architecture saving from adaptive rounds, both confirmed by per-stage profiling.

It is worth being explicit about why token selection does not translate into the speed-up one might expect. Tokenisation does not remove the cost of volumetric encoding, which dominates the front of the pipeline; the all-pairs structure scorer of Eq. (6) remains quadratic in token count even when the resulting messages are sparse; dense masked matrix operations do not run faster because many weights are near zero; and top-two routing with per-sample stopping saves wall-clock time only if kernels, token packing and batch scheduling actually exploit the skipped work. The 18,432-to-64 ratio is a 288:1 reduction in representation size, not a measured 288-fold reduction in floating-point operations, and it is easy to read the first as the second. Decomposing the measurement,

$$T_{\text{total}} = T_{\text{I/O}} + T_{\text{pre}} + T_{\text{enc}} + T_{\text{token}} + T_{\text{struct}} + \sum_{r=1}^{r_*} \left(T_{\text{graph},r} + T_{\text{expert},r} \right) + T_{\text{head}}, \tag{28}$$

shows which terms adaptive depth can affect at all: only the summation. A profile that does not separate these terms cannot attribute a saving to the mechanism that produced it, which is why the 104 ms figure appears here with no clinical interpretation attached.

A concrete profiling protocol follows in Section 11. Training energy is not estimated, because GPU-hours alone do not determine it without measured or explicitly assumed power draw and facility overhead.



All measurements on a single A100 80 GB. Latency is model-only forward time at batch size one and excludes DICOM loading, resampling and lung segmentation.

Figure 7: Computational profile. (Left) Cost relative to HAMLET; the parameter and memory advantages over the dense encoder coexist with higher latency than either comparator. (Centre) Distribution of reasoning depth across the test cohort. (Right) The depth–timing contrast. Latency values are provisional model-timing records with no documented measurement boundary.

11 Discussion

The most defensible reading of this evidence is that lesion representation, predictive confidence and computational allocation are related but separable design problems, and that a single architecture can address them with mechanisms that do measurably different work. The token and topology ablations produce the largest ranking changes. Replacing the evidential head barely moves ranking but moves calibration by a factor of two, and fixed-depth reasoning preserves discrimination while consuming almost twice the rounds. Presenting these as three distinct roles is a stronger scientific argument than claiming that every component improves every metric, and it is also what the numbers actually show.

The comparison hierarchy clarifies what is novel. The dense baseline asks whether an explicit lesion representation earns its complexity against volumetric fusion. The static graph asks the harder question of whether adaptive candidates and learned connectivity add anything once a relational model and an evidential head are already in place, and it is the more informative of the two because it holds the predictive objective fixed. The component replacements then localise plausible mechanisms. None of this establishes that the full system outperforms a protocol-matched state-of-the-art risk predictor; published image-only and multimodal models are already strong, and the next contribution has to be demonstrated under common conditions rather than inferred from mismatched published scores.

Multimodality is architecturally real in HAMLET — the clinical node conditions lesion processing before pooling. Architectural presence, however, is not the same as contribution. A model can contain a modality that adds nothing, and it can equally well exploit a clinical field that leaks outcome information through the timing of its collection. That is why the unresolved feature schema is a scientific gap rather than a reporting omission. Recovering the join, verifying input timestamps and quantifying image-only, clinical-only and joint performance are central to the paper's own claim. The clinical-only reference need not be elaborate: a regularised logistic model on the same permitted covariates would reveal how much outcome signal sits outside the images, and would make the multimodal gain interpretable for the first time.

The two error groups mark a coherent clinical boundary. Small peripheral nodules and diffuse ground-glass change are hard because the evidence is spread thinly across weak local contrast and broader context; fibrotic scars and vascular bifurcations create the opposite problem by producing strong local features that resemble malignancy. Neither failure mode looks like something a larger classifier fixes. Longitudinal comparison with prior screens, explicit vessel modelling and reader-provided phenotype labels are more plausible routes, and the first of these is also what distinguishes the strongest published image-only system [2].

Interpretability deserves the same empirical treatment. Attention maps and graph rollouts describe model computation; they are useful for debugging and they are not evidence that a radiologist would endorse the reasoning. A reader study could ask whether highlighted tokens contain the clinically relevant finding, whether graph edges correspond to relationships a radiologist recognises, and whether the vacuity score changes review behaviour. That would convert qualitative illustrations into a test of decision support, which is what the clinical claim ultimately requires.

Finally, a paper of this kind should be able to accommodate results that do not flatter the method, and several here do not. The MoE contribution is small and statistically unresolved. Retrieval costs latency for a negligible gain and its configuration status is ambiguous. HAMLET is slower than both internal comparators in the measured timings. External generalisation is unestablished, and the selective-prediction story is withdrawn until the uncertainty record is reconciled. These boundaries sharpen rather than weaken the contribution: if a matched rerun shows that a simpler configuration retains the important benefits, the simpler configuration should become the preferred model.

Priorities for the next study

The experiments below are ordered by how much each would change what the paper can claim.

1. **Isolate the modality contribution.** Train image-only, clinical-only and joint variants of the same backbone and

ablate the clinical node, quantifying missingness and temporal availability of each covariate. This is the experiment that decides whether "multimodal" is descriptive or load-bearing, and it is the first thing we intend to report.

2. **Strengthen the statistical protocol.** Report ROC, precision–recall and FROC curves alongside calibration with patient-clustered bootstrap intervals, prespecify the primary comparisons, and apply Holm's correction to the exploratory family [32].

3. **Remove architectural confounds.** With matched encoder, candidate count, head and edge budget, compare the learned graph against fixed spatial and density-matched topologies in a 2×2 token-by-graph factorial design. Compare the single-Gumbel formulation against the Binary Concrete sampler of Eq. (7), and report ε selection on validation data under a fixed stopping estimator.

4. **Complete the calibration and validation package.** Add validation-fitted temperature scaling to the softmax comparator; report Brier, NLL, adaptive-bin ECE, error-detection AUROC and risk–coverage curves with intervals; then evaluate an independent institutional cohort with frozen scores and a stated outcome horizon, checking participant overlap for any NLST-derived resource.

5. **Profile, then simplify.** Vary expert count over $\{1, 2, 4, 8\}$ and fixed against adaptive depth; record synchronised per-stage latency, median and 95th-percentile batch-one timings, actual active-token counts, and allocated and reserved memory, separating preprocessing and I/O from model time per Eq. (28). If the small MoE and retrieval gains do not replicate, prefer the simpler variant.

12 Limitations

Four limitations bound what this study shows. The evaluation cohort is outcome-enriched at 49.3% positive, so every threshold-dependent quantity in Section 6 is specific to that distribution and requires prevalence-aware recalibration before it means anything clinically. Calibration itself rests on a single ten-bin ECE; Brier score, negative log-likelihood and adaptive-bin estimates would characterise it better, and a temperature-scaled softmax comparator is the control that would establish how much of the evidential head's advantage survives cheap post-hoc calibration.

Second, the comparison is not fully capacity-matched. The dense encoder carries 312 M parameters against HAMLET's 104 M, and the fixed k-NN graph does not have the same edge budget as a learned graph at 25% density, so part of each margin may reflect capacity or connectivity volume rather than the mechanism under test. Third, intervals come from examination-level bootstraps; where participants contribute repeated screens, patient-clustered resampling would be the stricter procedure, and the reported intervals describe one trained model rather than variance across seeds. The p-values are unadjusted across the comparison family.

Fourth, several implementation choices remain to be characterised rather than merely described: edge direction and self-loop handling, whether edges are sampled or thresholded at inference, the sensitivity of the graph to the temperature schedule and sparsity coefficient, and the selection of the halting threshold. Beyond these, the label derives from confirmed cancer outcome and does not model intermediate or pre-malignant states, which plausibly adds noise around early adenocarcinoma; the token decoder's LUNA16 warm-start creates a dependence on a separate detection corpus whose effect at low candidate prevalence has not been isolated; and the objective combines five terms with two annealed schedules, so hyperparameter interactions are likely stronger than in the single-loss baselines. No prospective study addresses safety, workload or reader benefit, and nothing here should be read as evidence of deployment readiness.

Data and code availability. NLST data are available through NCI's Cancer Data Access System under its access conditions [33, 34, 35]; LIDC-IDRI and LUNA16 are established public resources [23, 22]. Model code, training and evaluation configurations, split manifests, checkpoint identifiers and score-level outputs for all baselines and ablations are available from the author on reasonable request, subject to the NLST data-use conditions that prevent redistribution of participant-level identifiers.

Ethics and disclosures. This work analyses de-identified public research data. NLST, LIDC-IDRI and LUNA16 are de-identified public research resources used under their respective access conditions.

13 Conclusion

HAMLET offers a coherent way to combine CT-derived lesion representations with clinical covariates inside an adaptive graph, and the internal comparisons report higher discrimination than both a dense multimodal reference and a static graph with an evidential head. The ablation pattern is the more interesting result. Most of the change in ranking is attributable to token representation and learned connectivity, whereas almost all of the change in calibration comes from the evidential objective. Adaptive depth changes neither, and affects only how much computation an examination consumes. Those are three differentiated hypotheses, each attached to a specific mechanism and a specific experiment, and that structure is what the paper contributes.

The boundaries of the claim are worth stating as plainly as the result. The evaluation cohort is enriched, so calibration and every threshold-dependent figure require prevalence-aware recalibration before they carry clinical meaning. The comparison is internal: a protocol-matched evaluation against published risk models, on a common split with a harmonised endpoint, remains the test that would place HAMLET in the wider literature. And an independent institutional cohort, with weights and thresholds frozen in advance, is what would establish generalisation. What the present study supports is narrower and, we think, more useful than a leaderboard position: lesion representation, predictive confidence and computational allocation are separable design problems, and a single architecture can address them with mechanisms whose contributions can be told apart.

Acknowledgment

The author thanks the NLST investigators and the LIDC-IDRI and LUNA16 organisers for maintaining de-identified research datasets and curated annotations.

References

[1] National Lung Screening Trial Research Team, "Reduced lung-cancer mortality with low-dose computed tomographic screening," *New England Journal of Medicine*, vol. 365, no. 5, pp. 395–409, 2011.

[2] D. Ardila, A. P. Kiraly, S. Bharadwaj, et al., "End-to-end lung cancer screening with three-dimensional deep learning on low-dose chest computed tomography," *Nature Medicine*, vol. 25, pp. 954–961, 2019.

[3] P. G. Mikhael, J. Wohlwend, A. Yala, et al., "Sybil: A validated deep learning model to predict future lung cancer risk from a single low-dose chest computed tomography," *Journal of Clinical Oncology*, vol. 41, no. 12, pp. 2191–2200, 2023.

[4] C. Niu, Q. Lyu, C. D. Carothers, et al., "Medical multimodal multitask foundation model for lung cancer screening," *Nature Communications*, vol. 16, p. 1523, 2025.

[5] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. CVPR*, 2017, pp. 2117–2125.

[6] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. ECCV*, 2020, pp. 213–229.

[7] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *Proc. ICLR*, 2018.

[8] L. Franceschi, M. Niepert, M. Pontil, and X. He, "Learning discrete structures for graph neural networks," in *Proc. ICML*, PMLR 97, 2019, pp. 1972–1982.

[9] E. Jang, S. Gu, and B. Poole, "Categorical reparameterization with Gumbel-Softmax," in *Proc. ICLR*, 2017.

[10] C. J. Maddison, A. Mnih, and Y. W. Teh, "The Concrete distribution: A continuous relaxation of discrete random variables," in *Proc. ICLR*, 2017.

[11] N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, and J. Dean, "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer," in *Proc. ICLR*, 2017.

[12] W. Fedus, B. Zoph, and N. Shazeer, "Switch Transformers: Scaling to trillion parameter models with simple and efficient sparsity," *Journal of Machine Learning Research*, vol. 23, no. 120, pp. 1–39, 2022.

[13] A. Graves, "Adaptive computation time for recurrent neural networks," arXiv:1603.08983, 2016.

[14] M. Sensoy, L. Kaplan, and M. Kandemir, "Evidential deep learning to quantify classification uncertainty," in *Advances in Neural Information Processing Systems*, vol. 31, 2018.

[15] Z. Han, C. Zhang, H. Fu, and J. T. Zhou, "Trusted multi-view classification with dynamic evidential fusion," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 45, no. 2, pp. 2551–2566, 2023.

[16] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[17] Y. Ovadia, E. Fertig, J. Ren, et al., "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[18] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. ICML*, PMLR 70, 2017, pp. 1321–1330.

[19] G. W. Brier, "Verification of forecasts expressed in terms of probability," *Monthly Weather Review*, vol. 78, no. 1, pp. 1–3, 1950.

[20] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[21] Y. Li and Y. Fan, "DeepSEED: 3-D squeeze-and-excitation encoder-decoder convolutional neural networks for pulmonary nodule detection," in *Proc. IEEE ISBI*, 2020, pp. 1866–1869.

[22] A. A. A. Setio, A. Traverso, T. de Bel, et al., "Validation, comparison, and combination of algorithms for automatic detection of pulmonary nodules in computed tomography images: The LUNA16 challenge," *Medical Image Analysis*, vol. 42, pp. 1–13, 2017.

[23] S. G. Armato III, G. McLennan, L. Bidaut, et al., "The Lung Image Database Consortium (LIDC) and Image Database Resource Initiative (IDRI): A completed reference database of lung nodules on CT scans," *Medical Physics*, vol. 38, no. 2, pp. 915–931, 2011.

[24] F. I. Tushar, A. Wang, L. Dahal, et al., "Reproducible benchmarking for lung nodule detection and malignancy classification across multiple low-dose CT datasets," arXiv:2405.04605, revised 2026.

[25] L. Klein, F. Isensee, K. Maier-Hein, et al., "LungEvaty: A scalable, open-source transformer-based deep learning model for lung cancer risk prediction in LDCT screening," arXiv:2511.20116, 2025.

[26] A. Hatamizadeh, V. Nath, Y. Tang, D. Yang, H. R. Roth, and D. Xu, "Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images," in *MICCAI Brainlesion Workshop*, 2022, pp. 272–284.

[27] J. Arevalo, T. Solorio, M. Montes-y-Gómez, and F. A. González, "Gated multimodal units for information fusion," arXiv:1702.01992, 2017.

[28] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2021.

[29] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, 2019.

[30] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson, "Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach," *Biometrics*, vol. 44, no. 3, pp. 837–845, 1988.

[31] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.

[32] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[33] National Cancer Institute, "NLST images and datasets," Cancer Data Access System, accessed 2026.

[34] National Cancer Institute, "NLST participant data dictionary," April 2022.

[35] National Cancer Institute, "NLST CT image info masterfile data dictionary," April 2022.